

This is an exciting and scalable premise. By focusing on the **core gameplay loop** and a **simpler execution to launch a Scratch game by the end of the year**, you're creating a foundation for a more complex game or expanded narrative later. Here's how to simplify and refine your concept while keeping it impactful and aligned with your broader goals:

1. Core Game Concept: Build and Sustain

Premise:

- **Tyler King**, a resourceful teenager, must transform his backyard into a **self-sustaining ecosystem** to support his family.
- Starting with **one resource (plastic recycling)**, players unlock new systems and skills, gradually advancing to higher levels of sustainability.

Simplified Gameplay for Scratch:

- **Goal:** Create a **choose-your-own-adventure-style resource management game** with clear objectives and incremental complexity.
 - **Focus:** Start with **one core mechanic (plastic recycling)** and build the game around that. Expand later.
 - **Levels:**
 1. **Plastic Recycling:** Collect bottles and turn them into 3D printing filament.
 2. **Energy Generation:** Use recycled materials to build solar panels or biogas systems.
 3. **Farming:** Build a FarmBot to grow food and sustain the household.
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2. Game Mechanics: Simplified Decision Tree

A. Resource Management

- **Resources:**
 - Collect materials like plastic bottles, scraps, and food waste.
 - Process these materials into usable components (e.g., filament, compost, biogas).

B. Decision Tree

- Every decision impacts the available resources and future opportunities.
- **Example:**

- Do you recycle plastic to create filament for selling or use it to build components for solar panels?
- Your choice determines which systems you can unlock later.

C. Skills and Upgrades

- Players unlock skills as they progress:
 - Level 1: Basic Recycling (turn bottles into filament).
 - Level 2: Build a FarmBot for food production.
 - Level 3: Automate systems with AI (interactive intelligence).
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3. Simplified Narrative

Story Framework:

- **Act 1:** Tyler's family faces financial struggles during the pandemic. He decides to build a self-sustaining backyard.
- **Act 2:** Tyler starts with plastic recycling and gradually builds systems for energy and food production.
- **Act 3:** His success inspires neighbors to adopt similar systems, creating a ripple effect in the community.

Key Elements:

- **Interactive Intelligence:**
 - A Raspberry Pi-powered AI assistant, gifted by his father, helps Tyler make decisions.
 - It provides hints, tracks progress, and offers story-based encouragement.
 - **Grandparents and Neighbors:**
 - Grandparents can assist with small tasks, like sorting materials.
 - Tyler can call on neighbors for additional resources or collaboration.
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4. Scratch Game Execution

Simplify Gameplay for Launch:

- **Core Gameplay:**
 - Focus on **one system (plastic recycling)** for the first version.
 - Add **decision-based interactions**: e.g., what to recycle, how to use resources, and when to upgrade.

- Create a **progress bar or status window** to show goals and achievements.
- **Design:**
 - Use **click-and-drag mechanics** for tasks like sorting materials or assembling parts.
 - Add simple animations to show progress (e.g., turning bottles into filament).

Structure:

1. **Level 1:** Collect plastic bottles and recycle them into filament.
 2. **Level 2:** Use filament to build a small solar panel for energy.
 3. **Level 3:** Unlock and build the FarmBot to start food production.
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5. Long-Term Potential

Expand the Game:

- Add systems for biogas production, composting, and water recycling.
- Introduce more complex decision trees with branching outcomes.

Narrative Growth:

- Expand Tyler's story to include the community and larger initiatives.
- Incorporate global connections (e.g., his father's work in Kenya).

Educational Modules:

- Use the game to teach **STEM skills** like recycling, IoT, and robotics.
 - Offer supplemental guides for real-world DIY projects.
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6. Path to Market

Immediate Goal:

- Develop a **Scratch prototype** focusing on plastic recycling and filament production.
- Launch the game by **end of the year** as a proof of concept.

Funding Opportunities:

- Apply for grants (e.g., 4Culture First Grant) to expand the game.
- Partner with libraries or schools to integrate the game into STEM education programs.

Marketing Strategy:

- Share the game on platforms like Scratch and Itch.io.
 - Highlight its educational value and sustainability themes.
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7. Final Thoughts

By focusing on **one core mechanic** for the Scratch game (e.g., plastic recycling) and integrating incremental upgrades, you can launch a meaningful project quickly. This serves as a proof of concept for the larger narrative and mechanics you want to explore later.